Cameroon Main Results

Cleaning diagnostics and NACAM employment elasticities

April 10, 2026

Roadmap

1. BDF Cleaning Results
2. NACAM Elasticity Results

Data cleaning overview

- The working panel is organized at the firm-year level, so duplicate handling changes both sample size and identification.
- Economic sector is based on the NACAM variable, which according to documentation its based on a Regional harmonized classifier linked to ISIC.
- The cleaning note isolates the two steps that most affect downstream interpretation: duplicate resolution and the NACAM-to-ISIC crosswalk.

Firm-Year Duplicate Cleaning

- Working key: `firmid` × `fin_yr`
- Perfect duplicates would be collapsed with original row order as the tie-breaker.
- Conflicting duplicate firm-years are fully removed from the cleaned analysis file, while we figure out the best approach for handling them.

	Value
Original observations	9,325
Perfect duplicate firm-years	0
Rows dropped from perfect duplicates	0
Conflicting duplicate firm-years	127
Rows dropped from conflicting duplicates	257
Final cleaned observations	9,068

NACAM-to-ISIC Crosswalk

- Observed legacy NACAM codes are audited directly from the Cameroon BDF file.
- ISIC Rev.4 divisions are assigned only when the official mapping is one-to-one.
- Ambiguous branches remain flagged for manual review rather than being forced into a noisy division code.
- Since the NACAM is already based on the ISIC structure, the analysis file retains the original NACAM variable for elasticity estimation and reporting, while the ISIC variable is used for sectoral aggregation and cross-country comparisons.

	Value
Observed legacy NACAM codes covered	37
Codes with exact ISIC Rev.4 division assignment	17
Codes flagged for manual review	20

Cleaning Takeaways

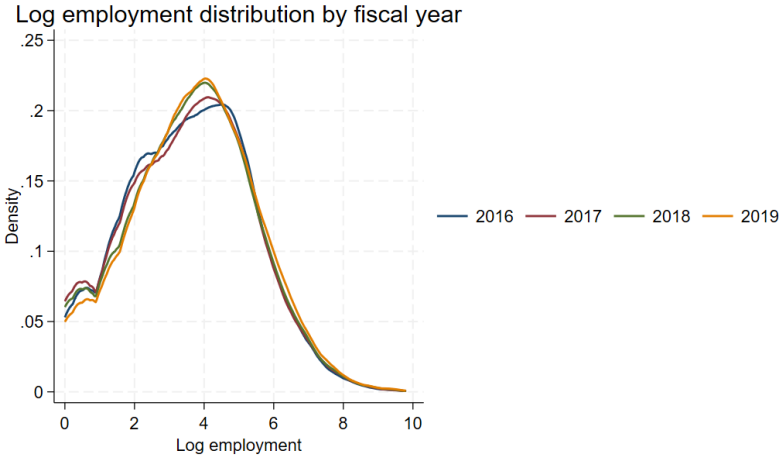
- The cleaned panel falls from 9,325 to 9,068 observations after removing duplicate firm-years.
- There are 127 conflicting firm-years, accounting for 257 dropped obs. Those observations are excluded rather than resolved ad hoc.
- The crosswalk covers 37 observed legacy NACAM codes, but only 17 map cleanly to a single ISIC Rev.4 division.
- To avoid introducing noise we keep the analysis using NACAM structure.
- The remaining 20 codes are explicitly flagged for manual review, which keeps sector harmonization transparent before pooled industry comparisons.

Specification and Sample

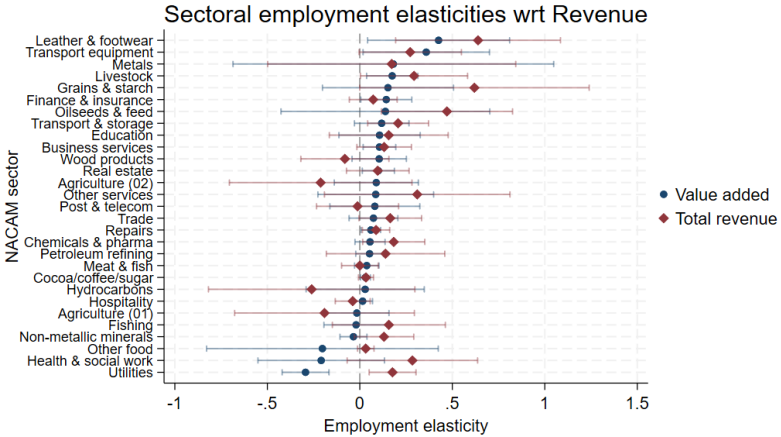
$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s \left(\ln(X_{it}) \times D_{is}\right) + \varepsilon_{it} \tag{1}$$

- The unit of observation is a firm-year (i, t).
- E_{it} is employment for firm i in year t.
- X_{it} is either value added or total revenue, estimated in separate regressions.
- D_{is} equals 1 if firm i belongs to NACAM sector s, and 0 otherwise.
- Sector is treated as a firm characteristic, so each firm contributes to one NACAM interaction term.
- α_i are firm fixed effects and δ_{st} are sector-by-year effects.
- β_s is the elasticity for firms in sector s, identified through the interaction between $\ln(X_{it})$ and the sector indicator D_{is}.
- Standard errors clustered at the firm level.
- Inclusion rule for reported sectors: at least 30 observations and 10 firms in both outcome samples.

Employment Distribution by Fiscal Year

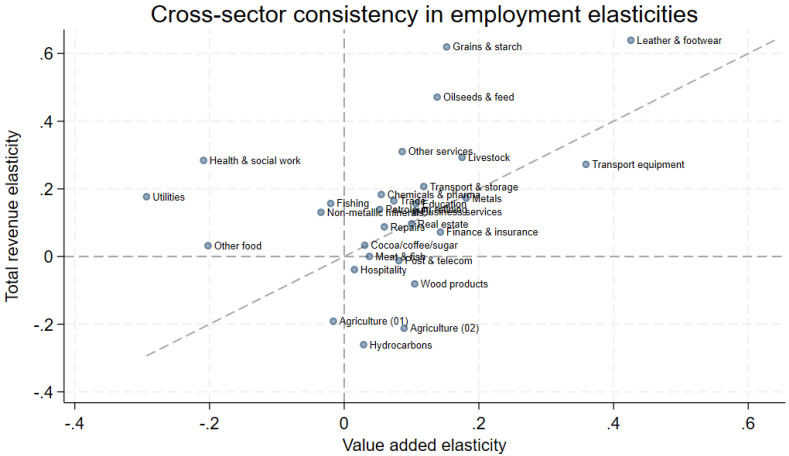


Main Elasticity Estimates Across NACAM Sectors



Sectors shown meet the minimum firm and observation thresholds in both models.

Cross-Model Consistency



Selected Sectors From the VA Ranking

- Table shows the top three and bottom two sectors by value-added elasticity.
- Total-revenue elasticities are included as a cross-check on whether the signal is consistent across revenue concepts.

	VA_Elasticity	TotRev_Elasticity	VA_Obs	TotRev_Obs
NACAM 17	0.426	0.639	148	164
NACAM 27	0.359	0.272	125	145
NACAM 24	0.181	0.172	45	53
NACAM 41	-0.209	0.284	243	263
NACAM 29	-0.294	0.177	54	56

Main Takeaways

- Elasticities vary meaningfully across NACAM sectors rather than lining up around a single Cameroon-wide coefficient.
- Several sectors show positive employment responses under both value added and total revenue, with NACAM 17 standing out most clearly.
- Negative or weak value-added elasticities remain concentrated in a small set of sectors, which is useful for prioritizing follow-up diagnostics.
- The cleaning and harmonization steps are not bookkeeping only; they determine which firm-years and sector codes enter the elasticity analysis.